

Problem-Solution Fit

Date	27 June 2026
Team ID	SWTID-2026-7769
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure that the proposed Credit Card Approval Prediction System effectively addresses the needs, challenges, and expectations of its target users. The canvas helps identify the key customer problems, their underlying causes, existing limitations, proposed solutions, and the value delivered by the system. Fill out the canvas in the respective sections following the original visual template while aligning each element with the project's objectives and machine learning-based credit card approval process.

1. CUSTOMER SEGMENT(S) [CS] - Individuals applying for credit cards - Bank customers seeking instant eligibility checks - Financial institutions and banks - Credit officers evaluating applications	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Limited knowledge of approval criteria - Long waiting time for approval - Incomplete financial understanding - Dependence on manual banking processes	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Existing Solutions: Manual verification, traditional banking systems, rule-based approval systems Pros: - Widely used - Reliable for standard cases - Easy to implement Cons: - Time-consuming - Prone to human errors - Limited prediction accuracy - Lack of intelligent decision support
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Delays in credit card approval process (Very Frequent) - Difficulty in assessing applicant eligibility (Frequent) - Manual verification consumes significant time (Frequent) - Inconsistent approval decisions due to human error (Occasional)	9. PROBLEM ROOT / CAUSE [RC] - Manual evaluation of applications - Complex eligibility requirements - Human errors in decision-making - Lack of automated prediction systems	7. BEHAVIOR + ITS INTENSITY [BE] - Checks eligibility before applying (High) - Compares different credit card options (Medium) - Frequently follows up on application status (Medium)
3. TRIGGERS TO ACT [TR] - Applying for a new credit card - Need for quick eligibility verification - Banks processing large numbers of applications - Requirement to reduce approval time 4. EMOTIONS BEFORE / AFTER [EM] Before: - Anxious - Uncertain - Frustrated - Confused After: - Confident - Informed - Satisfied - Relieved	10. YOUR SOLUTION [SL] Credit Card Approval Prediction System uses Machine Learning to predict credit card approval accurately and efficiently. - Collects applicant demographic and financial information - Performs data preprocessing and feature engineering - Predicts approval using the Random Forest model - Generates a detailed risk audit report - Stores prediction history for future analysis - Reduces manual effort and improves decision accuracy	8. CHANNELS OF BEHAVIOR [CH] ONLINE - Bank Websites - Mobile Banking Applications - Financial Service Portals - Online Credit Card Platforms OFFLINE - Bank Branches - Customer Service Centers - Financial Consultants - Banking Help Desks